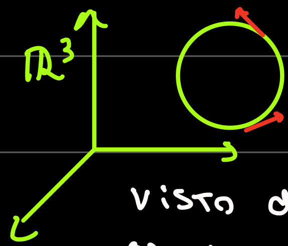
Relatividade Geral

Introduction

Paralelismo:

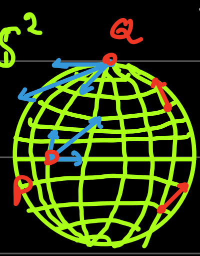
$\mathbb{R}^3$



visto do $\mathbb{R}^3$ os vetores não são paralelos

$\vec{V} = \lambda \vec{W}$

S^2



Já de S^2 eles são paralelos

"Conexión" equivalente matemática

bases dependentes

$\vec{A} = 2e_1 + 3e_2$ X

$\vec{A}_p = 2e_1(P) + 3e_2(P)$

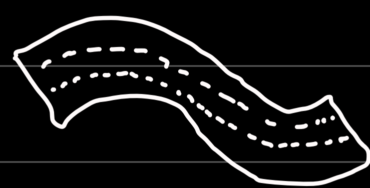
$\vec{A} = \underbrace{2e_1 + 3e_2}$

dependen de onde se ve?

$\{T_pM\}$

||

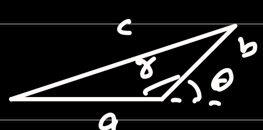
Tangent Bundle



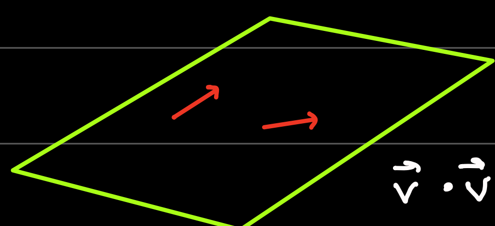
Metric $\longleftrightarrow$ Scalar Product

(linear, simétrico) em Euclides

Lei dos Cosenos



$\mathbb{R}^2$

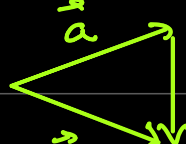


v

$\vec{v} \cdot \vec{v} = v^2$

$c^2 = a^2 + b - 2ab\cos\gamma$

$\cos(\pi - \theta) = -\cos\theta$ $\quad c^2 = a^2 + b^2 + 2ab\cos\theta$



$\vec{c} = \vec{a} + \vec{b}$

$\vec{c} \cdot \vec{c} = \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + 2|\vec{a}||\vec{b}|\cos\theta$

$$(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b}) = \vec{a}\cdot\vec{a} + \vec{b}\cdot\vec{b}$$

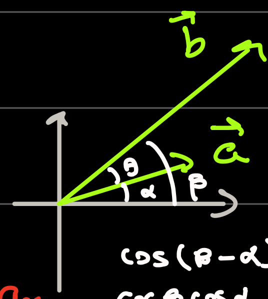
$$a^2 + \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{a} + b^2 = a^2 + b^2 + 2|\vec{a}||\vec{b}|\cos\theta$$

$$2\vec{a}\cdot\vec{b} = 2ab\cos\theta$$

Em Geometria Euclidiana $\rightarrow$ $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$

Teorema

$$\vec{a}\cdot\vec{b} = a_x b_x + a_y b_y$$



Tudo isso para espaço Euclidiano

$\cos(\beta-\alpha) =$
$\cos\beta\cos\alpha + \operatorname{sen}\beta\operatorname{sen}\alpha$

$$\underbrace{ab(\cos\beta\cos\alpha + \operatorname{sen}\beta\operatorname{sen}\alpha)}_{\vec{a}\cdot\vec{b}}$$

a_x, a_y, b_x, b_y

Isso tudo só é verdade se

$$\vec{a} = a_x\hat{e}_x + a_y\hat{e}_y$$
$$\vec{b} = b_x\hat{e}_x + b_y\hat{e}_y$$

$$\begin{cases} \hat{e}_x\cdot\hat{e}_x = 1 \\ \hat{e}_y\cdot\hat{e}_y = 1 \\ \hat{e}_1\cdot\hat{e}_2 = 0 \end{cases}$$

$$(\alpha e_0 + \beta e_1)\cdot(\gamma e_1 + \delta e_2)$$

$$\alpha \underbrace{e_0\cdot e_0}_{1} + \alpha\delta \underbrace{e_1 e_0}_{0} + \beta\gamma \underbrace{e_0\cdot e_1}_{0} + \beta\delta \underbrace{e_1\cdot e_1}_{1}$$

What if it Doesn't

$$\mathbb{R}^2 \{e_1, e_2\} \quad e_i e_j = \delta_{ij}$$

$g_{11}=1$ $g_{12}=g_{21}=0$ $g_{22}=1$

$$\{e_1+e_2, e_2\} \equiv \{u_1, u_2\} \quad \text{base}$$

$\vec{A} = 2u_1 + 3u_2 \qquad \vec{B} = u_1 - u_2$

Qual é o produto escalar agora?

$$\vec{A}\cdot\vec{B} = (2u_1 + 3u_2)\cdot(u_1 - u_2)$$

$$= 2\underbrace{u_1\cdot u_1}_{g_{11}} - 2\underbrace{u_1 u_2}_{g_{12}} + 3\underbrace{u_2 u_1}_{g_{21}} - 3\underbrace{u_2 u_2}_{g_{22}}$$

Métrica

$$\begin{cases} g_{11} = u_1 u_1 = (e_1 + e_2)\cdot(e_1 + e_2) = 1 + 1 = 2 \\ g_{12} = u_1 u_2 = (e_1 + e_2)e_2 = 1 \\ g_{21} = 1 \\ g_{22} = 1 \end{cases}$$

nova métrica

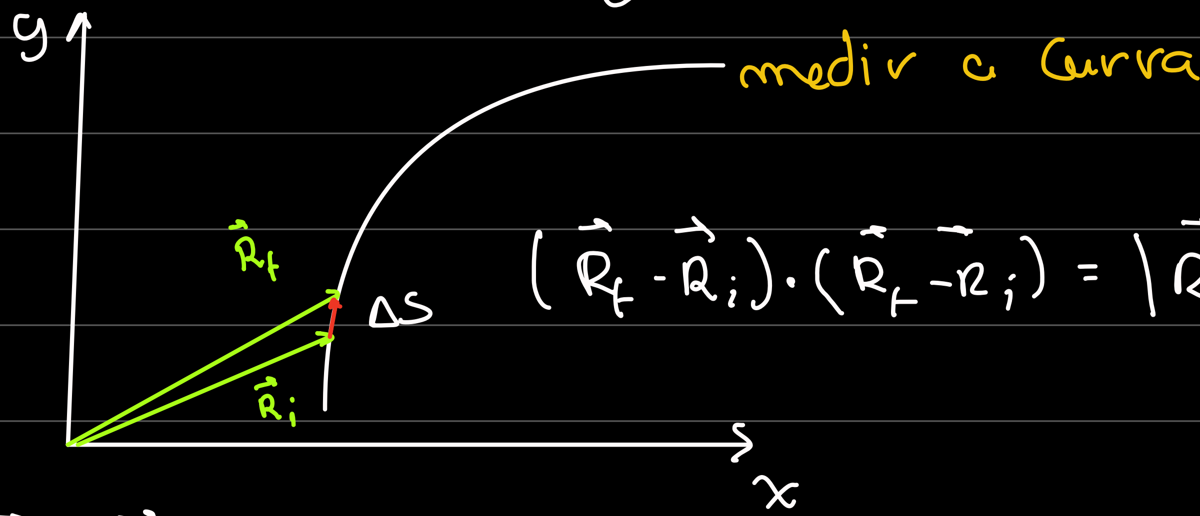
$$\vec{A} = a_1 u_1 + a_2 u_2 \qquad \vec{B} = b_1 u_1 + b_2 u_2$$

$$\vec{A}\cdot\vec{B} = a_1 b_1 g_{11} + a_1 b_2 g_{21} + \dots$$

$\{e_1, e_2\}$ $\quad g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\{u_1, u_2\}$ $\quad g_{ij} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$

$$\vec{A}\cdot\vec{B}=\sum\sum a_i b_j g_{ij}$$



$$(\vec{R}_f-\vec{R}_i)\cdot(\vec{R}_f-\vec{R}_i)=|\vec{R}_f-\vec{R}_i|^2$$

$$\vec{R}_f-\vec{R}_i = x_f e_1 + y_f e_2 - (x_i e_1 + y_i e_2)$$

$$=(x_f-x_i)e_1+(y_f-y_i)e_2$$

$$\Delta s = (\Delta x e_1 + \Delta y e_2)(\Delta x e_1 + \Delta y e_2)$$

$$\Delta s^2 = (\Delta x)^2 g_{11} + 2\Delta x \Delta y\, g_{12} + \Delta y^2 g_{22}$$

$$\lim_{f\to i}(\Delta s)^2 \to (ds)^2 = g_{11}\,dx^2 + 2dx\,dy\,g_{12} + g_{22}\,dy^2$$

$$ds=\sqrt{g_{11}dx^2+2g_{12}dx\,dy+g_{22}dy^2}$$

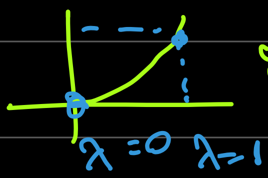
$$S=\int_a^b ds = \int_a^b \left(g_{11}dx^2+2g_{12}dx\,dy+g_{22}dy^2\right)^{\frac{1}{2}} \frac{d\lambda}{d\lambda}$$

Parametrizar a curva

λ

$\lambda = \dots$

$$\int_{\lambda=0} \left(g_{11}\frac{dx^2}{d\lambda^2} + 2g_{12}\frac{dx}{d\lambda}\frac{dy}{d\lambda} + g_{22}\frac{dy^2}{d\lambda^2}\right)^{\frac{1}{2}} d\lambda$$



$y = x^2$ $\quad$ $x = \lambda$ $\quad$ $\lambda = 0$

así $y = \lambda^2$

$$\frac{dx}{d\lambda} = 1 \qquad \frac{dy}{d\lambda} = 2\lambda$$

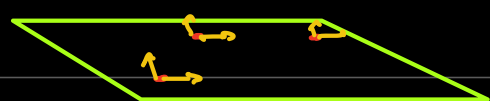
$$\int_{\lambda=0}^{\lambda=1} \left(g_{11} + 2g_{12}\cdot 2\lambda + g_{22}\cdot 4\lambda^2\right)^{\frac{1}{2}} d\lambda$$

Para métrica de Base Canónica $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\int_0^1 (1 + 4\lambda^2)^{\frac{1}{2}} d\lambda = 1.4879$$

Métrica $\vec{v}\cdot\vec{v} = v^2$ → linear
↳ Simétrica

Intro II



(Base Dual de $\{e_1, e_2\}$) $\equiv \{e^1, e^2\}$

$$\left.\begin{aligned} e^1\cdot e_1 &= 1 \\ e^1\cdot e_2 &= 0 \\ e^2\cdot e_1 &= 0 \\ e^2\cdot e_2 &= 1 \end{aligned}\right\}$$ → Sempre independente da métrica

$$e^1 = a e_1 + b e_2$$

$$e^2 = c e_1 + d e_2$$

$$(a e_1 + b e_2)\cdot e_1 = a g_{11} + b g_{21} = 1$$

$$(a e_1 + b e_2)\cdot e_2 = a g_{12} + b g_{22} = 0$$

$$\ldots$$

$$c g_{11} + d g_{21} = 0$$

$$c g_{12} + d g_{22} = 1$$

$$\begin{cases} a g_{11} + b g_{21} = 1 & (\cdot g_{12}) \\ a g_{12} + b g_{22} = 0 & (\cdot g_{11}) \end{cases}$$

$$\begin{cases} a g_{12} g_{11} + b g_{12} g_{21} = g_{12} \\ - \\ a g_{12} g_{11} + b g_{11} g_{22} = 0 \end{cases}$$

$$\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}$$

$$b(\underbrace{g_{12} g_{21} - g_{11} g_{22}}_{-\det(g)}) = g_{12}$$

$$b = -\frac{g_{12}}{\det(g)} \; ; \quad a = \frac{g_{22}}{\det(g)}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{\det(g)} \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix} = g_{ij}^{-1} \equiv g^{ij}$$

$$g^{ij} = \begin{pmatrix} \frac{g_{22}}{|g|} & -\frac{g_{12}}{|g|} \\ -\frac{g_{21}}{|g|} & \frac{g_{11}}{|g|} \end{pmatrix}$$

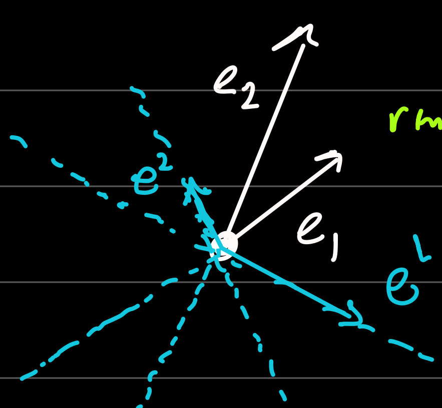
Therefore Dual Basis

einstein notation

$$\left.\begin{aligned} e^1 &= g^{11} e_1 + g^{12} e_2 \\ e^2 &= g^{21} e_1 + g^{22} e_2 \end{aligned}\right\} \quad e^i \quad i = 1, 2$$

$$\left\{\begin{aligned} & e^i = \sum_j g^{ij} e_j \rightsquigarrow \underbrace{e^i = g^{ij} e_j}_{\text{einstein summation convention}}, \\ & \underbrace{e^i \cdot e_j = \delta^i_j}_{*} \quad \text{Dual Basis definition} \end{aligned}\right.$$

$\{e_1, e_2\}$ $\{e^1, e^2\}$



remember

$e^1 \cdot e_1 = 1$

$e^1 \cdot e_2 = 0$

Covariant, Contravariantes Vectors

ex $\quad g_{ij} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad \{e_1, e_2\}$ $\qquad g_{ij} \neq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \{e_1, e_2\} \neq$ canonica

$$g^{-1}_{ij} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \Rightarrow g^{ij} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$e^1 = \frac{2}{3}e_1 - \frac{1}{3}e_2$$

$$e^2 = -\frac{1}{3}e_1 + \frac{2}{3}e_2$$

$$\underbrace{-\frac{2}{3}e_1 + \frac{4}{3}e_2}_{} = 2e^2$$

$$e^2 = e^1 + 2e^2$$

$$\vec{A} = 2e_1 + 3e_2$$

$$= 2(g_{11}e^1 + g_{12}e^2) + 3(g_{21}e^1 + g_{22}e^2)$$

$$= (2g_{11} + 3g_{21})e^1 + (2g_{12} + 3g_{22})e^2$$

$$= 7e^1 + 8e^2$$

$$e_2 = g_{21}e^1 + g_{22}e^2 = e^1 + 2e^2$$

Dual $\rightarrow$ Normal

$$e_i = g_{ij}e^j$$

$$\vec{A} = \underbrace{2e_1 + 3e_2}_{\text{Base da Métrica}} = \underbrace{7e^1 + 8e^2}_{\text{base dual}}$$

agora a Nomenclatura

$$\vec{A} = 2e_1 + 3e_2 = 7e^1 + 8e^2$$

A^1 A^2 — Componentes de $\vec{A}$ Contravariantes

A_1 A_2 — Componentes de $\vec{A}$ Covariantes

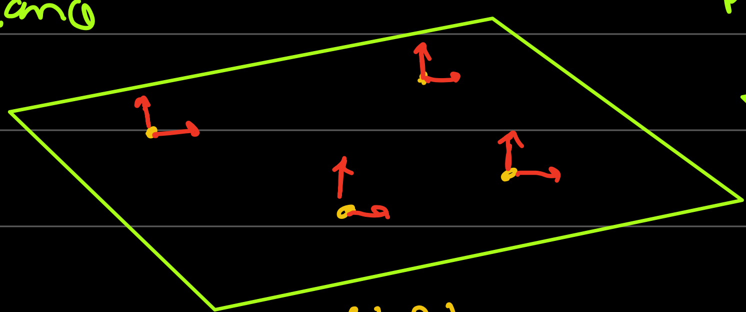
$$\vec{A} = A^i e_i = A_i e^i$$

mesmo objeto em bases diferentes

Métrica

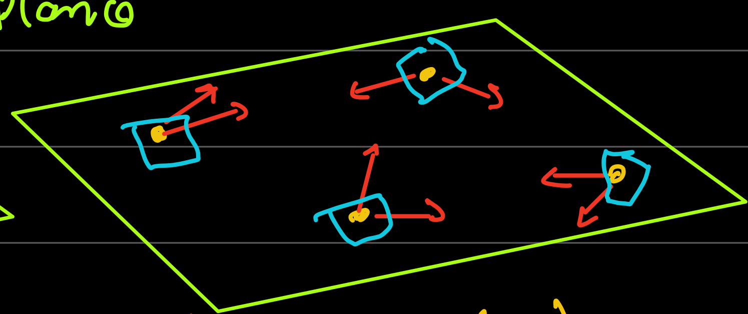
$$g_{ij} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$ não necessariamente constantes

Plano



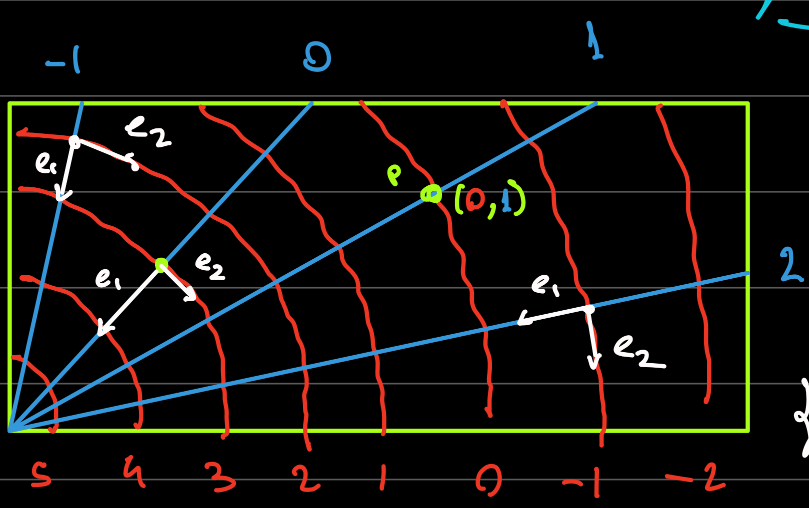
$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Plano



métrica depende do ponto

$\square \Rightarrow$ espaços tangentes são os mesmos (plano)



g_{ij} (Ponto)

$\{e_1, e_2\}$ Base Coordenada